3 Simulation: Repaired and Completed

Scope: the simulation contribution only. The empirical examples in the original group report are outside this chapter. The table and figure numbering 1-6 follows the original simulation presentation; the estimates are not copied from the old report.

3.1 Setup and Objective

We examine two questions: whether a Neyman-orthogonal IV-type score reduces nuisance-estimation bias relative to a non-orthogonal ML ratio, and whether out-of-sample nuisance predictions reduce training-data reuse bias. Random Forest, XGBoost, and Lasso are evaluated under both linear and nonlinear DGPs.

$$Y_i = \theta_0 D_i + g_0(X_i) + U_i, \qquad D_i = m_0(X_i) + V_i$$

The true effect is 0.5, p=20, and X is Gaussian with Toeplitz covariance $\mathrm{Cov}(X_j, X_k)=0.7^{|j-k|}$. U and V are mutually independent standard Gaussian disturbances, independent of X. The repaired orthogonalization bridge uses n=500, K=2, R=300. The cross-fitting comparison reuses the completed v6 baseline n=300, K=2, R=200 receipts without new fitting.

3.2 Data Generating Processes

The two experiments use different baseline coefficients. This distinction was missing from the original report: its unified DGP description matched the old Figure 2 code, but not the archived Figure 1 code. Here X_1 and X_3 correspond to Python columns 0 and 2; s(t)=1/(1+exp(-t)).

Experiment / DGP	m0(X)	g0(X)
Orthogonalization / linear	0.9 X1 + 0.20 X3	0.25 X1 + 0.25 X3
Orthogonalization / nonlinear	0.9 X1 + 0.20 s(X3)	s(X1) + 0.25 X3
Cross-fitting / linear	1.2 X1 + 0.35 X3	0.40 X1 + 0.40 X3
Cross-fitting / nonlinear	1.2 X1 + 0.35 s(X3)	1.2 s(X1) + 0.40 X3

3.3 Estimation and Reporting

Correct nuisance targets

The outcome regression and structural nuisance are different objects. In particular, learning E[Y|X] directly and then calling it structural g does not yield the claimed Neyman-orthogonal IV score. The repaired definitions are:

$$m_0(X) = E[D \mid X], \quad g_0(X) = E[Y - \theta_0 D \mid X]$$

$$\ell_0(X) = E[Y \mid X] = g_0(X) + \theta_0 m_0(X)$$

For each training fold, a joint Lasso pilot is fitted using D, all raw X, and sigmoid(X), with alpha=0.001. In the orthogonalization experiment, g is learned from the training response Y-pilot*D. In the reuse experiment, ell is learned from Y and g_hat=ell_hat-pilot*m_hat. The pilot, scalers, and nuisance models never use the held-out outcomes. True functions are used only for simulation generation and post-fit diagnostics.

$$\widehat{V}_i = D_i - \widehat{m}_i, \quad \hat{\theta}_{IV} = \frac{\sum_i \widehat{V}_i (Y_i - \hat{g}_i)}{\sum_i \widehat{V}_i D_i}$$

$$\hat{\theta}_{ML} = \frac{\sum_i D_i (Y_i - \hat{g}_i)}{\sum_i D_i^2}$$

The naive and IV estimators in Tables 1-2 share exactly the same held-out structural-g predictions. Their difference is the score, not different nuisance training or added pressure. The IV-type score is used throughout; no partialling-out estimator is restored.

Nuisance learners and controlled prior information

Treatment models use fixed candidate columns [0,2,4,6], including two inactive baseline candidates. RF and XGBoost treatment models use positive monotonicity constraints; treatment Lasso uses positive coefficients. Outcome RF/XGBoost use all raw covariates; nonlinear outcome Lasso uses a fixed raw/sigmoid dictionary. These are disclosed controlled priors, not unrestricted variable discovery.

RF: 300 trees; treatment leaf size 10, unrestricted depth, all candidate features; structural-g depth 8/leaf 2/sqrt features; ell unrestricted depth/leaf 1. XGBoost: treatment 300 trees/depth 1/rate .05/lambda 5; g 300/depth 3/rate .01; ell 600/depth 6/rate .1. Lasso alpha: treatment .001; g .02 linear/.06 nonlinear; ell .0001. All settings are inherited unchanged from the current mainline.

3.3 Estimation and Reporting (continued)

Uncertainty and histogram definitions

$$\psi_i = \widehat{V}_i(Y_i - \hat{g}_i - \hat{\theta}_{IV} D_i), \quad \widehat{SE} = \sqrt{\frac{\overline{\psi_i^2}}{n(\overline{\widehat{V}_i D_i})^2}}$$

Bias is the Monte Carlo average of theta_hat-0.5. MC SD is the sample standard deviation of the R estimates. Mean SE averages the per-replication sandwich SE. Coverage is the fraction of intervals theta_hat +/- 1.96*SE containing 0.5. MAE and RMSE are retained in the machine-readable summaries but are not conflated with absolute mean bias.

Figures 1-2 use raw errors theta_hat-0.5. Both panels in a learner row share a zero-centered normal reference whose SD equals that learner's DML MC SD, matching the original visual convention. Figures 5-6 use studentized errors (theta_hat-0.5)/SE and a standard-normal reference. Every observed error is included; paired panel axes have identical bounds.

Figures 3-4 and Tables 3-4 reproduce the original mean +/- 1.96*MC SD display. This is an approximate sampling spread of individual estimates, not a confidence interval for the Monte Carlo mean and not the per-replication causal confidence interval. Monte Carlo uncertainty about mean bias is reported separately as mean bias +/- $t_{R-1,.975}$*MC SD/sqrt(R).

What this bridge does and does not replicate

The n=500 experiment restores the old sample size and repetition count, and follows the archived Figure 1 DGP code. It changes the historical mismatched nuisance target to the repaired structural IV target and retains the current controlled learner settings. Therefore it is a repaired legacy-size bridge, not a literal reproduction of the old biased procedure or its printed numerical values.

A separate prespecified seed was fixed before observing any bridge result. All six learner-by-DGP cells and both methods are reported. No seed, learner, repetition count, or hyperparameter was selected using causal-effect accuracy. The server software preflight checked finiteness, identical model-bank inputs, absence of prediction overlap, and the algebraic error decomposition; low bias or a method win was not a gate.

3.3.1 Orthogonalization and Bias Reduction

The following tables preserve the original layout and show the repaired n=500, R=300 results. Interpret the size and uncertainty of the remaining bias separately from its reduction relative to non-orthogonal ML.

Table 1: Non-Orthogonal vs. IV-Type Estimation (Linear DGP, n=500, R=300)

Learner	Non-Orthogonal ML			IV-type DML		
	Bias	MC SD	Mean SE	Bias	MC SD	Mean SE
Random Forest	0.0523	0.0365	0.0324	0.0045	0.0439	0.0469
XGBoost	0.0378	0.0394	0.0329	0.0010	0.0463	0.0487
Lasso	0.0065	0.0430	0.0322	-0.0044	0.0433	0.0461

Table 2: Non-Orthogonal vs. IV-Type Estimation (Nonlinear DGP, n=500, R=300)

Learner	Non-Orthogonal ML			IV-type DML		
	Bias	MC SD	Mean SE	Bias	MC SD	Mean SE
Random Forest	0.0470	0.0414	0.0337	0.0042	0.0485	0.0474
XGBoost	0.0364	0.0446	0.0342	0.0027	0.0504	0.0483
Lasso	0.0301	0.0458	0.0332	0.0022	0.0480	0.0456

Bias is signed. Neither small displayed bias nor failure to detect a nonzero bias proves exact finite-sample unbiasedness. Learner-specific residual bias is retained, rather than removed by post-result tuning.

A small bias-variance tradeoff is present even in this bridge: for linear Lasso, absolute mean bias decreases from 0.00652 to 0.00436, but MAE changes from 0.034659 to 0.034771 and RMSE from 0.043406 to 0.043454. These small observed changes are not evidence of a statistically established deterioration, but they rule out describing every error metric as uniformly improved.

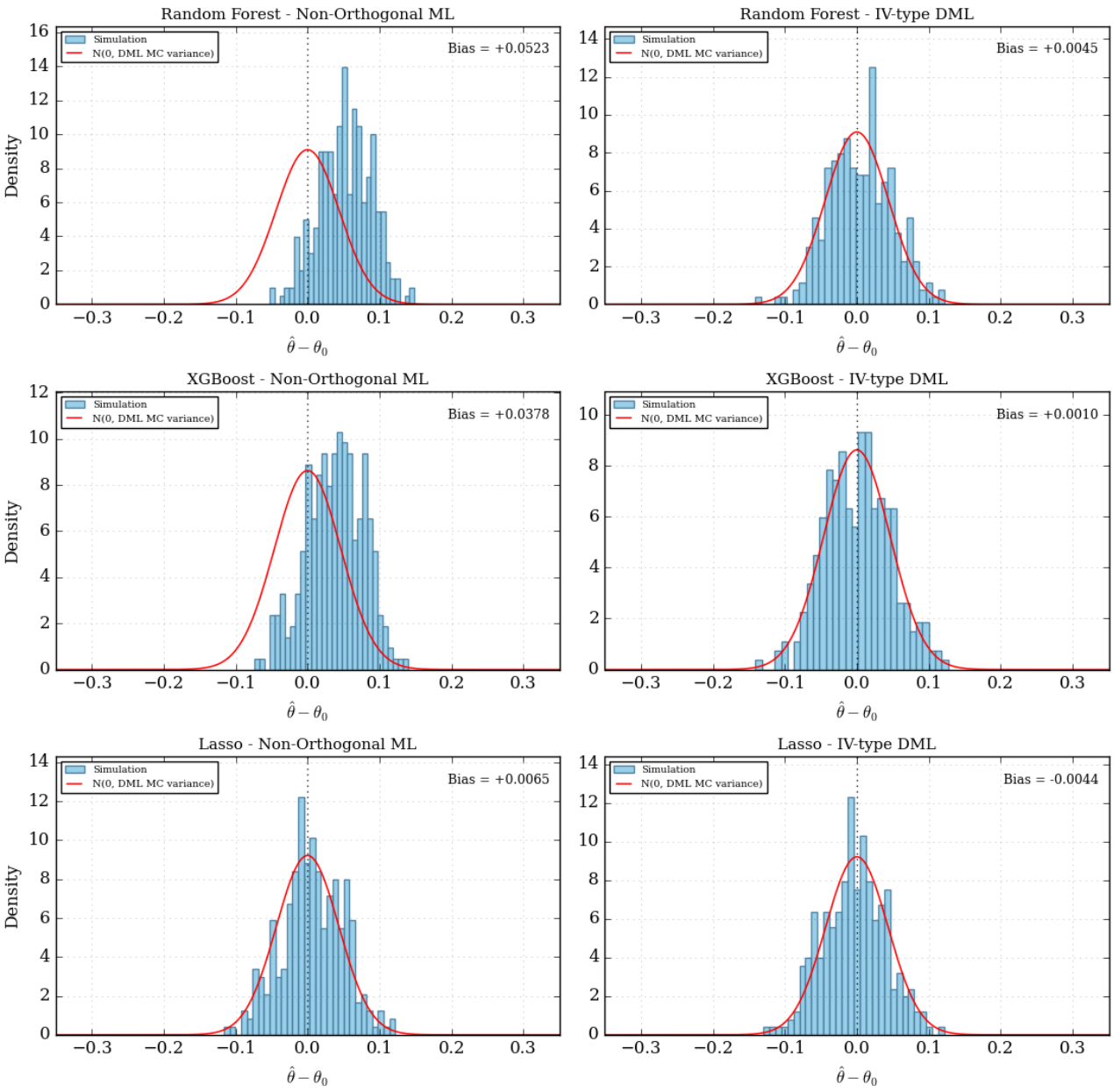


Figure 1: Linear DGP. Raw estimation errors: non-orthogonal ML versus repaired IV-type DML. n=500, p=20, R=300.

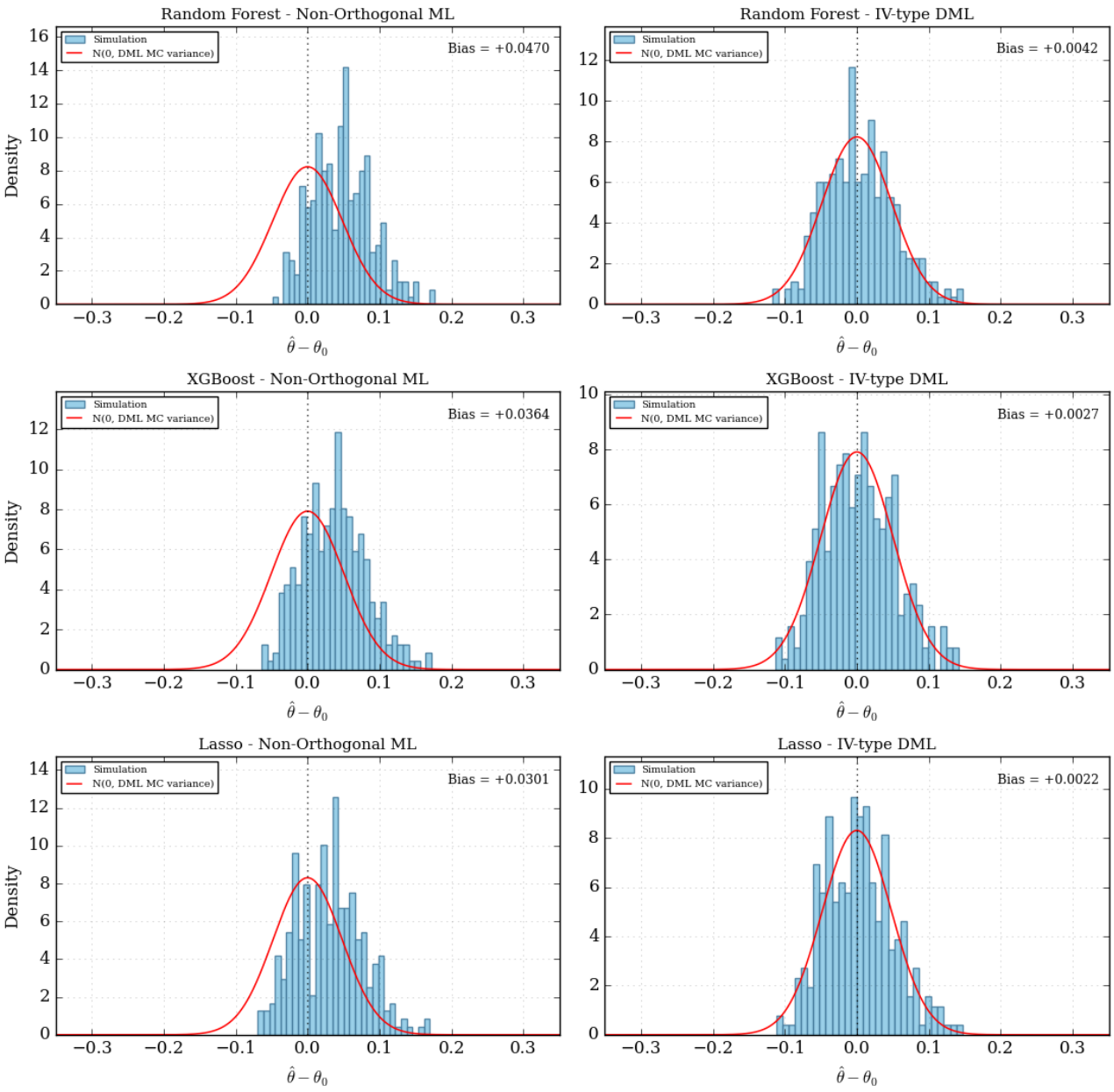


Figure 2: Nonlinear DGP. Raw estimation errors: non-orthogonal ML versus repaired IV-type DML. n=500, p=20, R=300.

3.3.2 Sensitivity to Learner Choice

Table 3: Learner Sensitivity (Linear DGP, n=500, R=300)

Learner	Mean estimate	Bias	MC SD	95% MC spread	Coverage
Random Forest	0.5045	0.0045	0.0439	[0.4184, 0.5905]	96.0%
XGBoost	0.5010	0.0010	0.0463	[0.4103, 0.5917]	95.3%
Lasso	0.4956	-0.0044	0.0433	[0.4108, 0.5805]	96.7%

Table 4: Learner Sensitivity (Nonlinear DGP, n=500, R=300)

Learner	Mean estimate	Bias	MC SD	95% MC spread	Coverage
Random Forest	0.5042	0.0042	0.0485	[0.4091, 0.5993]	94.7%
XGBoost	0.5027	0.0027	0.0504	[0.4038, 0.6016]	94.0%
Lasso	0.5022	0.0022	0.0480	[0.4080, 0.5964]	95.0%

The interval column is labelled MC spread to correct the ambiguity of the original 95% CI heading. Coverage still uses the individual sandwich-SE intervals. Differences in MC SD describe variance and cannot by themselves establish which learner is uniformly more accurate.

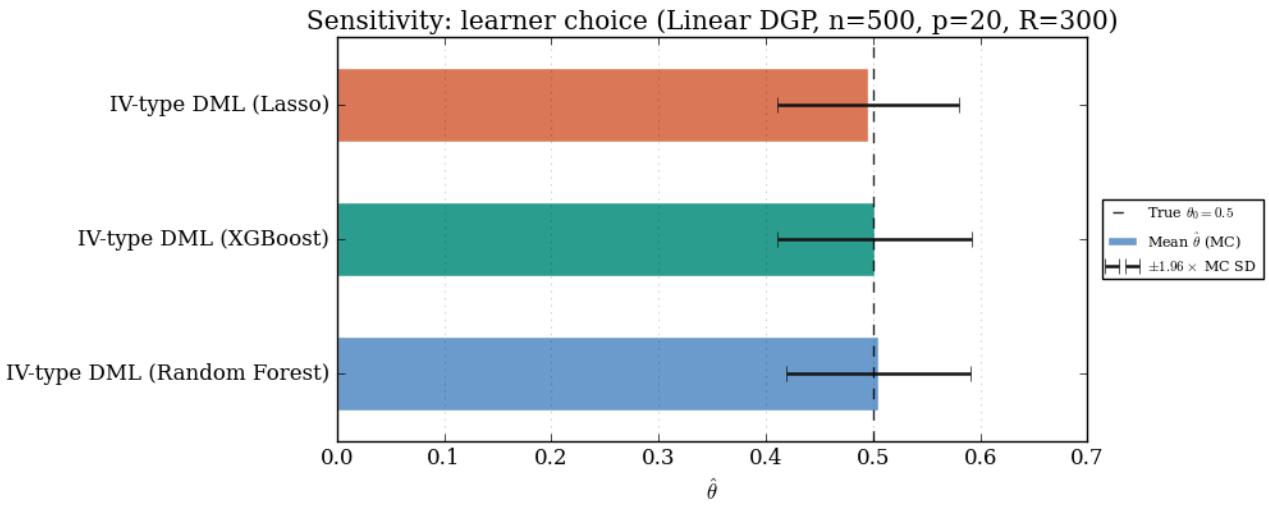


Figure 3: Sensitivity of repaired IV-type estimates to learner choice under the linear DGP. Error bars show +/- 1.96*MC SD, not the MC-mean confidence interval.

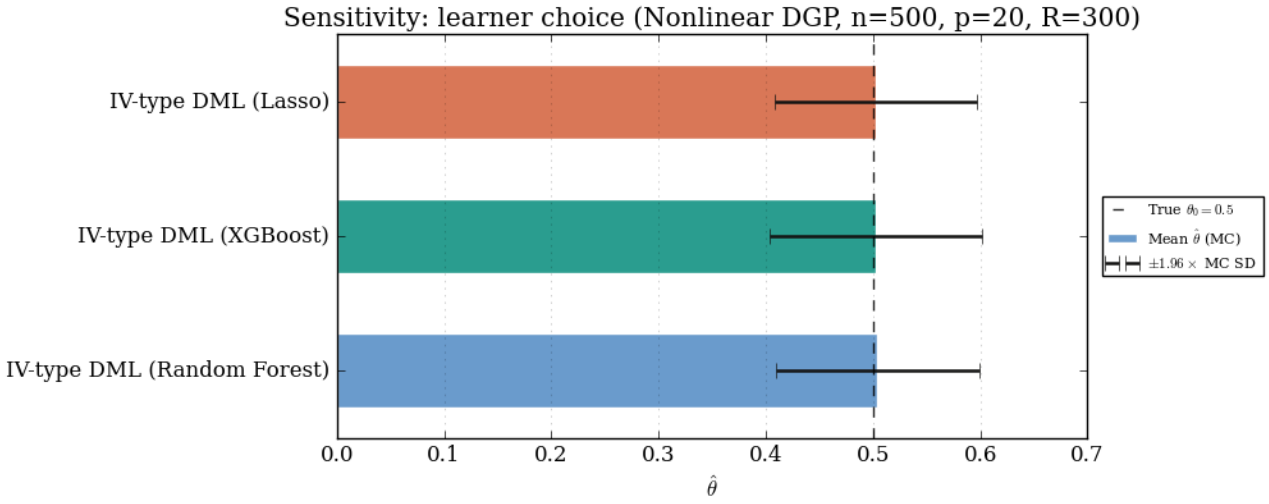


Figure 4: Sensitivity of repaired IV-type estimates to learner choice under the nonlinear DGP. Error bars show +/- 1.96*MC SD, not the MC-mean confidence interval.

3.3.3 Cross-Fitting and Training-Data Reuse

Table 5: Full-Sample vs. Cross-Fitting (Linear DGP, n=300, R=200)

Learner	Full-Sample No Split			IV-type DML		
	Bias	MC SD	Mean SE	Bias	MC SD	Mean SE
Random Forest	-0.2707	0.0253	0.0334	0.0107	0.0586	0.0592
XGBoost	-0.4858	0.0048	0.0435	-0.0116	0.0663	0.0732
Lasso	-0.0587	0.0536	0.0545	0.0041	0.0761	0.0712

Table 6: Full-Sample vs. Cross-Fitting (Nonlinear DGP, n=300, R=200)

Learner	Full-Sample No Split			IV-type DML		
	Bias	MC SD	Mean SE	Bias	MC SD	Mean SE
Random Forest	-0.2902	0.0261	0.0341	-0.0019	0.0665	0.0634
XGBoost	-0.4898	0.0036	0.0389	-0.0135	0.0777	0.0727
Lasso	-0.0562	0.0555	0.0549	0.0066	0.0717	0.0725

These are the existing v6 baseline n=300/K=2/base results, recomputed directly from 400 hash-verified receipts. No n=300 model was refitted. Full no-split trains on n observations, whereas each cross-fit model trains on n/2. This conventional contrast changes both prediction membership and training size; Tables 7-8 provide the separate same-bank control.

Cross-fitting substantially reduces the displayed bias but does not make every setting unbiased: linear RF retains +0.01069, linear XGBoost -0.01164, and nonlinear XGBoost -0.01349 mean bias. These cells are reported without retuning or replacing the completed experiment.

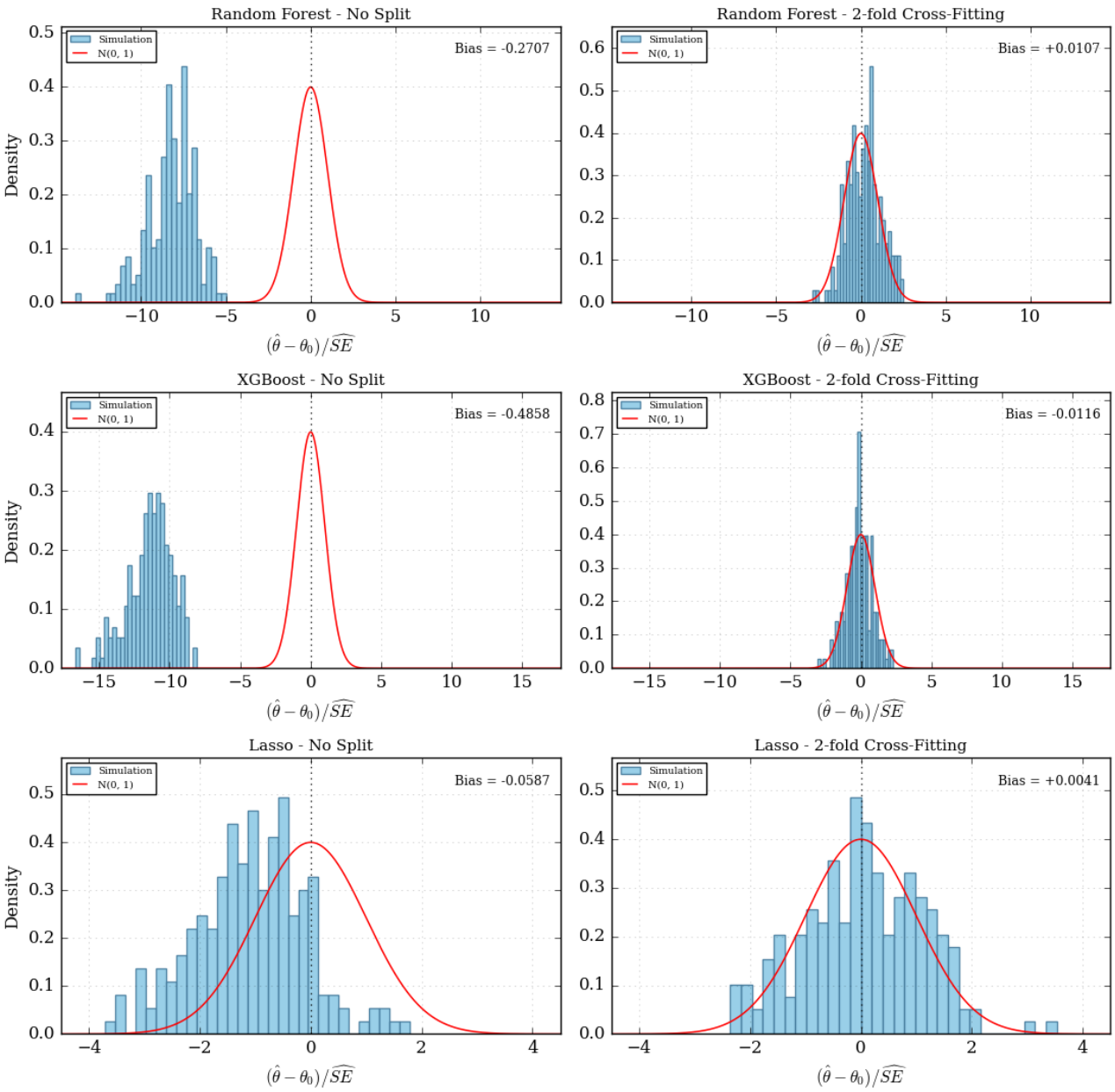


Figure 5: Linear DGP. Full-sample no-split versus 2-fold cross-fitting; studentized estimation errors. n=300, p=20, R=200.

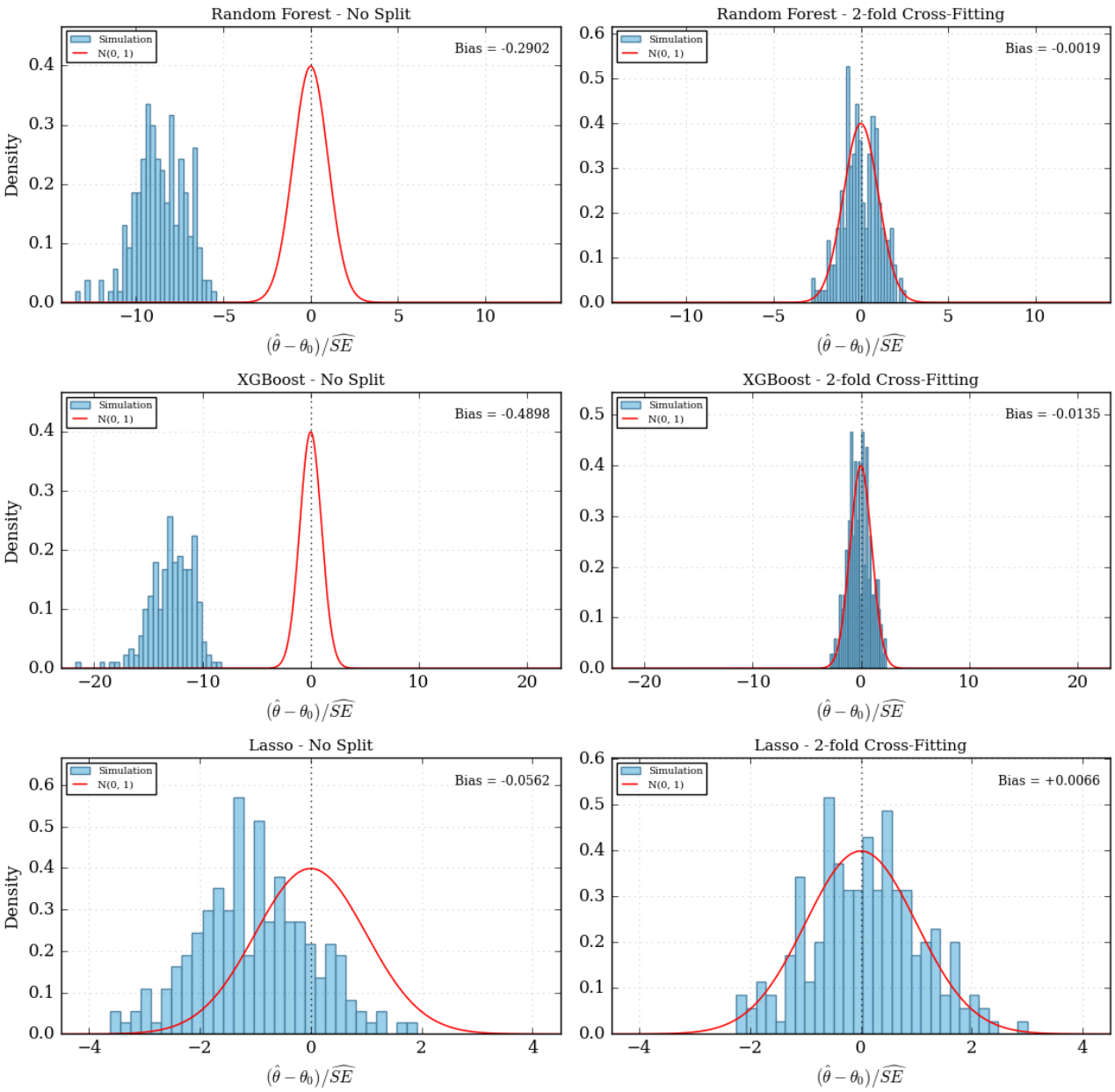


Figure 6: Nonlinear DGP. Full-sample no-split versus 2-fold cross-fitting; studentized estimation errors. n=300, p=20, R=200.

3.4 Controlled Extensions

3.4.1 Equal-training-size, shared-model-bank control

The same two fitted nuisance model banks predict every observation. Cross-fitting selects a model trained on the other half; reuse selects the model trained on that observation's own half. Training sizes, fitted models, targets, pilot settings, and seeds are identical. No artificial own-response pressure is introduced only into no-split.

Table 7: Equal-Training Reuse vs. Cross-Fitting (Linear DGP, n=300, R=200)

Learner	Equal-Training Reuse			IV-type DML		
	Bias	MC SD	Mean SE	Bias	MC SD	Mean SE
Random Forest	-0.2553	0.0278	0.0324	0.0107	0.0586	0.0592
XGBoost	-0.4939	0.0030	0.0473	-0.0116	0.0663	0.0732
Lasso	-0.1233	0.0499	0.0526	0.0041	0.0761	0.0712

Table 8: Equal-Training Reuse vs. Cross-Fitting (Nonlinear DGP, n=300, R=200)

Learner	Equal-Training Reuse			IV-type DML		
	Bias	MC SD	Mean SE	Bias	MC SD	Mean SE
Random Forest	-0.2784	0.0279	0.0339	-0.0019	0.0665	0.0634
XGBoost	-0.4948	0.0021	0.0416	-0.0135	0.0777	0.0727
Lasso	-0.1196	0.0546	0.0524	0.0066	0.0717	0.0725

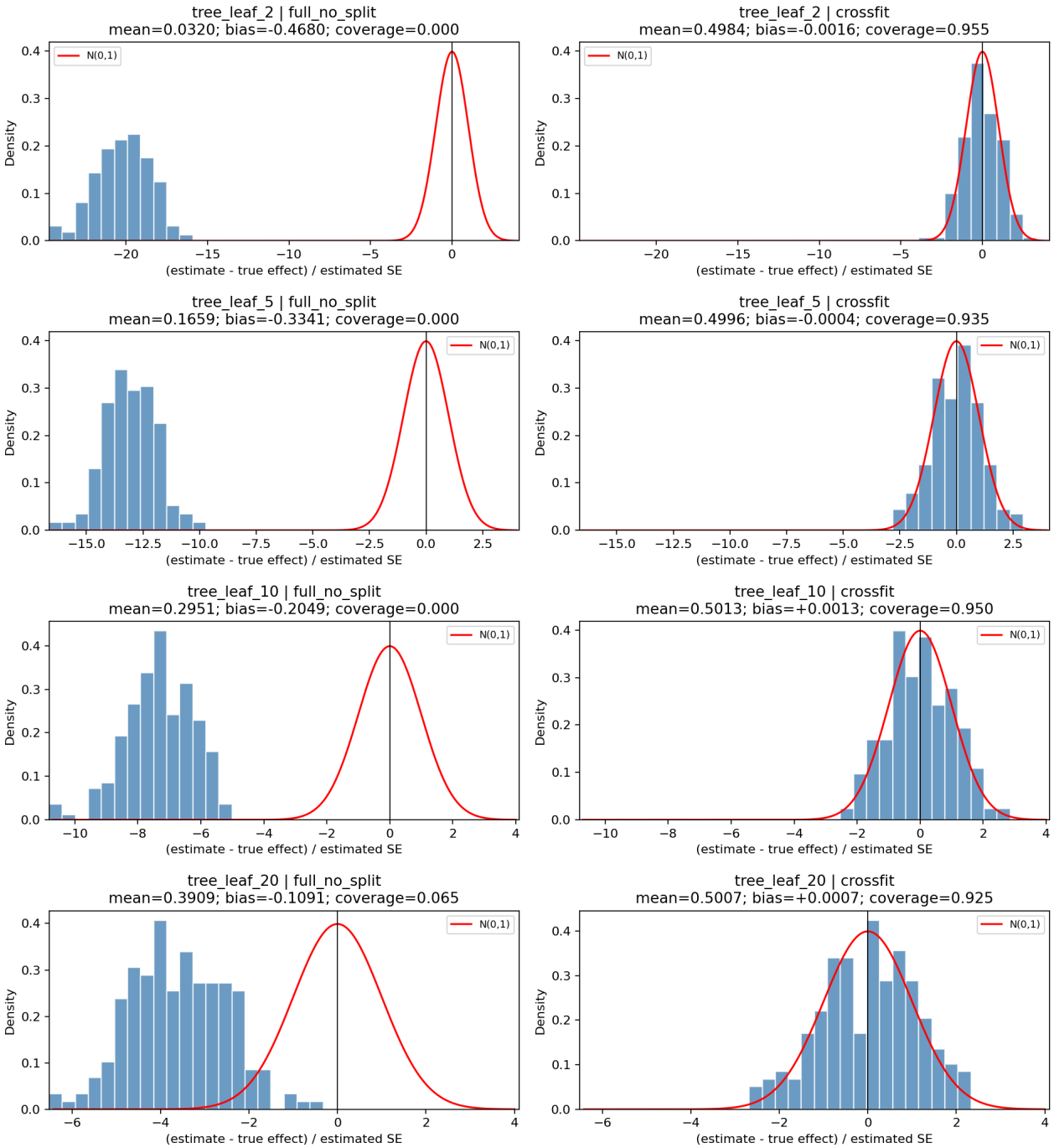
This controlled comparison isolates membership in the nuisance training sample more directly than full-sample no-split. Its results remain separate from the conventional full-sample comparison.

3.4.2 Correctly specified treatment channel

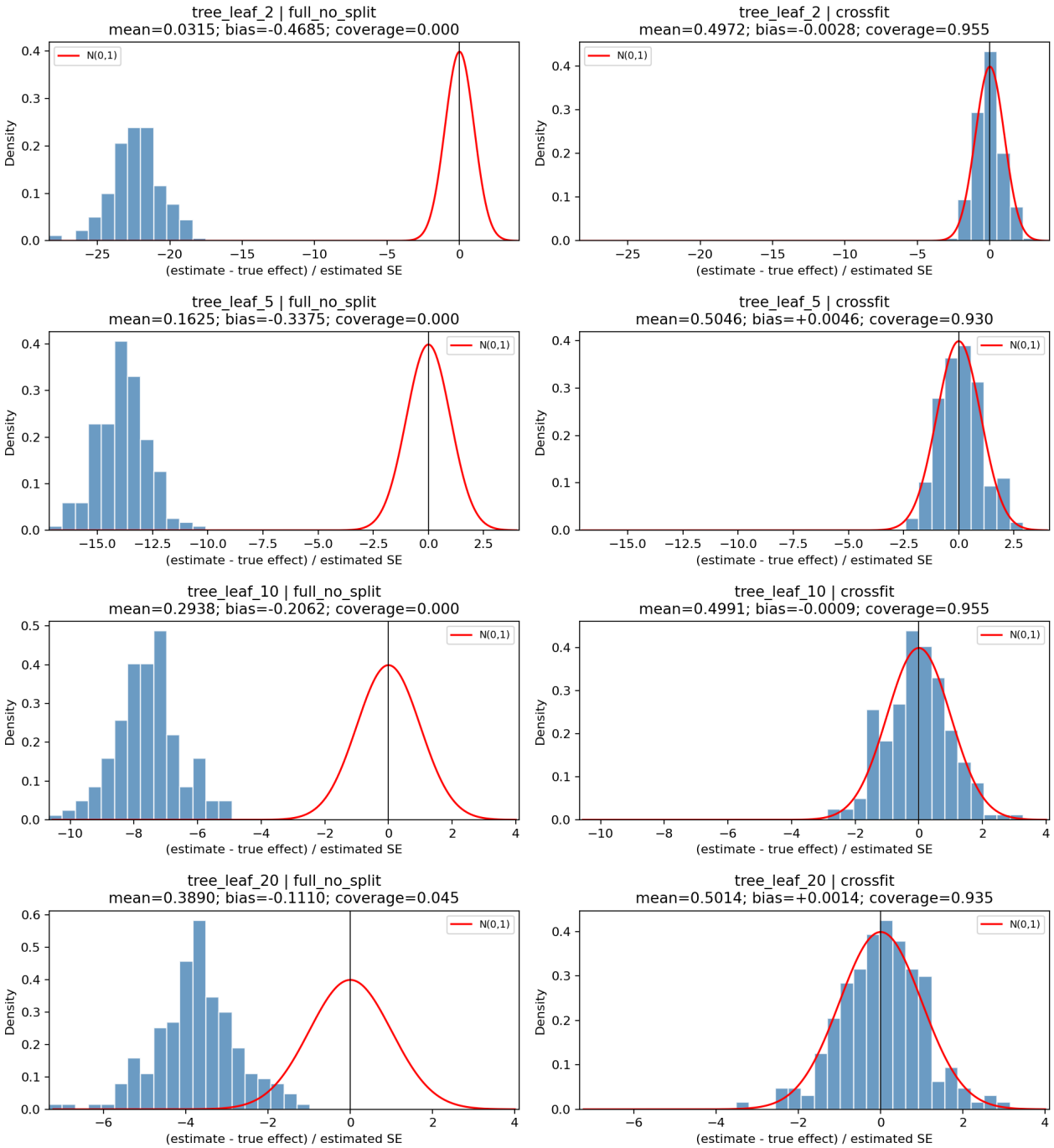
A separate completed n=1,000 study uses an unregularized Series OLS treatment model on [1, X1, X3, sigmoid(X1), sigmoid(X3)] and decision-tree outcome learners with minimum leaf sizes 2, 5, 10, and 20. It isolates the reuse mechanism in a correctly specified nuisance channel and is not substituted for the RF/XGBoost/Lasso experiments.

DGP / leaf	Mean estimate	Bias	MC SD	Mean SE	Coverage
linear / 2	0.4984	-0.0016	0.0518	0.0492	95.5%
linear / 5	0.4996	-0.0004	0.0477	0.0447	93.5%
linear / 10	0.5013	0.0013	0.0417	0.0406	95.0%
linear / 20	0.5007	0.0007	0.0413	0.0378	92.5%
nonlinear / 2	0.4972	-0.0028	0.0466	0.0487	95.5%
nonlinear / 5	0.5046	0.0046	0.0440	0.0442	93.0%
nonlinear / 10	0.4991	-0.0009	0.0405	0.0401	95.5%
nonlinear / 20	0.5014	0.0014	0.0375	0.0370	93.5%

Maximum absolute MC mean bias across these eight cross-fit settings is 0.00457467. This is a point-bias summary, not proof that every setting's entire bias-uncertainty interval lies inside +/-0.01. In particular, the nonlinear leaf-5 setting does not pass that stronger interval criterion.



Completed controlled Series OLS / decision-tree study: linear DGP. Existing source figure, unchanged.



Completed controlled Series OLS / decision-tree study: nonlinear DGP. Existing source figure, unchanged.

3.4.3 Sample Size, Folds, and Hyperparameters

The completed sensitivity study contains 600 reported summary cells: n=300/1,000/3,000, K=2/5, baseline and moderate DGP levels, and four one-nuisance-role-at-a-time parameter alternatives at n=1,000/K=2. It is not a full factorial hyperparameter grid. The complete JSON retains all adverse and inconclusive cells, paired MAE/MSE contrasts with the original Holm corrections, and nuisance-error decompositions.

Some reported settings are aliases rather than independent experiments: full-sample no-split does not depend on K, and the RF Figure 2 weak-outcome preset equals the baseline ell leaf-size setting. The count 600 is a count of reported summary cells, not 600 independent experimental configurations.

The exact moderate additions below apply to each experiment's own baseline nuisance functions (superscripts M and B denote moderate and baseline). Baseline and moderate comparisons share simulation draws. The Lasso raw/sigmoid dictionary does not contain the nonlinear interaction, providing a disclosed limitation rather than an arbitrary-misspecification guarantee.

$$m_0^M = m_0^B + 0.20X_5 + 0.15X_7, \quad g_0^M = g_0^B + 0.15X_5 + 0.15X_7$$

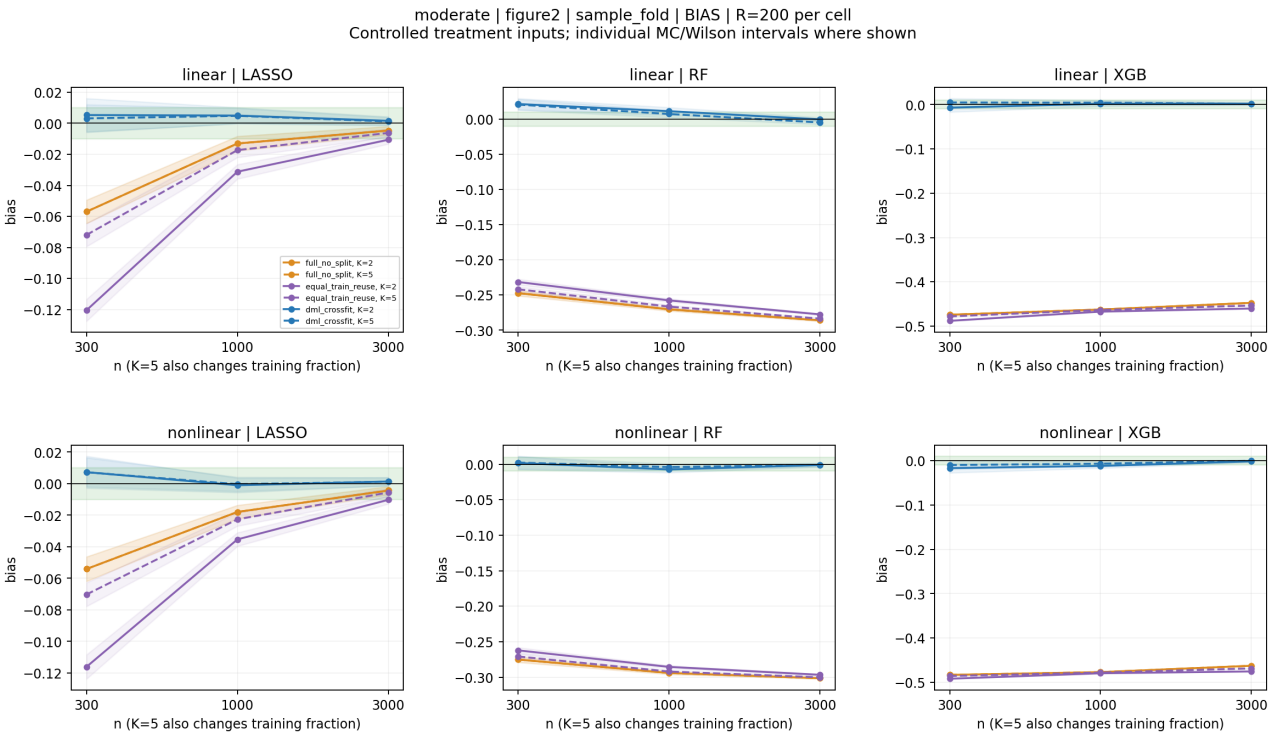
$$m_0^M = m_0^B + 0.20X_5 + 0.20s(X_7)$$

$$g_0^M = g_0^B + 0.25s(X_5) + 0.20X_7 + 0.30\tanh(X_1)\tanh(X_5)$$

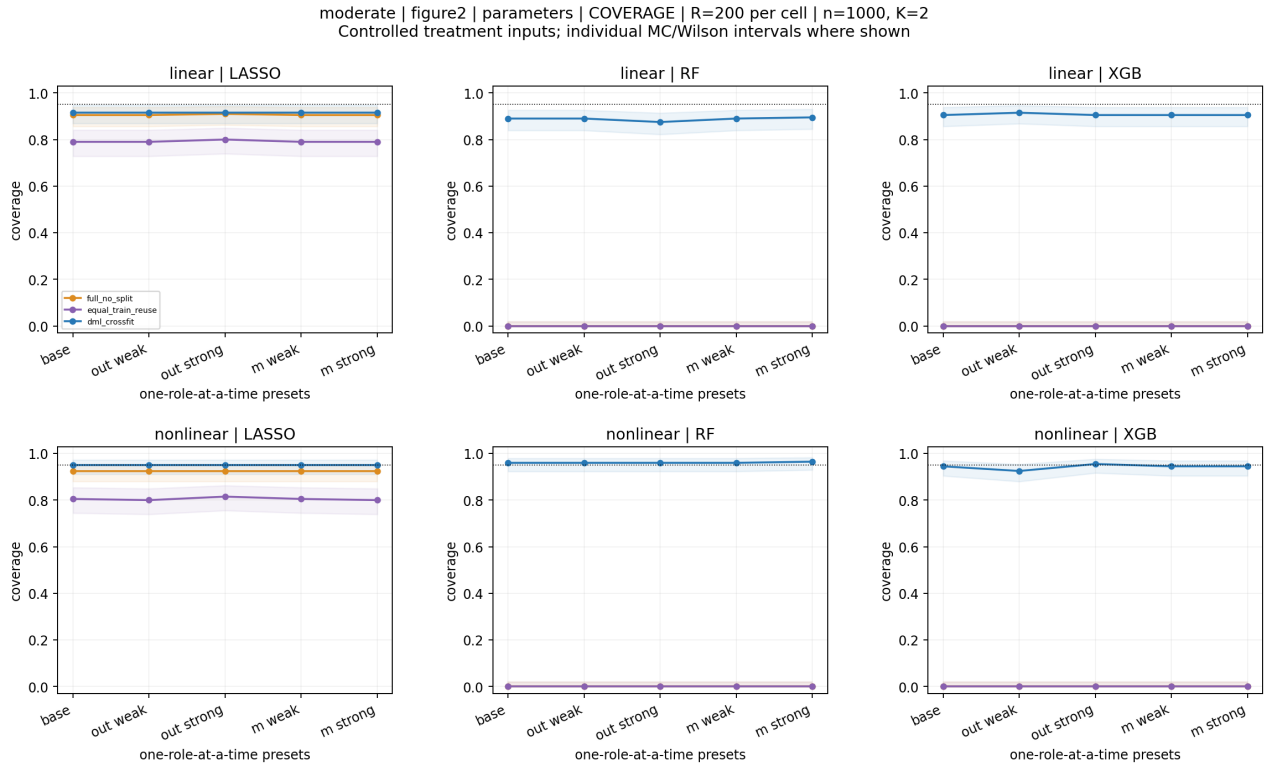
At the moderate n=1,000/K=2/base anchor, the equal-weight average absolute MC mean bias over six learner-by-DGP cells falls from 0.0365543 to 0.00387464 for non-orthogonal ML versus DML (89.4%), and from 0.255777 to 0.00648017 for full no-split versus cross-fitting (97.5%). These percentages are specific to that anchor, not averages over the whole sensitivity grid or MAE reductions.

Important adverse findings are retained: moderate linear RF Figure 1 DML bias is +0.01275; moderate linear Figure 2 RF bias is +0.01151; nonlinear Figure 2 XGBoost bias is -0.01209. Cross-fit coverage in the sensitivity grid can fall to 87.5%. In the moderate linear Lasso reuse example, bias decreases but MAE and RMSE increase. Five folds are not uniformly better than two.

The accompanying HTML gallery includes every existing selected-stage sensitivity figure, all 600 tabulated cells, all six restored primary figures, and all eight restored comparison/sensitivity tables. Complete numerical summaries are included without changing the source values.



Existing sensitivity evidence: moderate figure2 sample fold bias. Layout-only source figure.



Existing sensitivity evidence: moderate figure2 parameters coverage. Layout-only source figure.

3.5 Interpretation, Limitations, and Provenance

The experiments evaluate bias reduction, training-data reuse, variance, and interval calibration. They do not establish universal finite-sample unbiasedness, uniformly lower total error, or valid coverage for every learner. Histograms provide visual distribution diagnostics, not a formal proof of asymptotic normality. The known candidate treatment features, monotonicity restrictions, and correctly specified Series OLS control are explicit scope restrictions.

The original report's empirical bonus-policy and smoking examples are not rerun or altered. No IRM experiment or partialling-out estimator is added to the simulation contribution. Historical adverse ordinary-learner results remain in the existing nuisance-comparison summary, alongside the controlled results.

Reproduction boundary

New model fitting occurred only on the authorized Linux server, in one launch with 12 CPU workers and no GPU use. The new protocol fixed n=500, R=300, K=2, two DGPs, three learner families, and one seed before outcomes. Every one of the 600 jobs and 3,600 estimates is retained. The actual server exit was zero. Local work reads saved estimates and produces diagnostics, figures, tables, and this PDF; it imports no ML learner libraries.

The n=300 tables were independently recomputed from the original v6 receipts and match its saved summaries to numerical precision. Figure 1 bridge nuisance bank hashes match between naive and DML. Figure 2 equal-training reuse and cross-fit share fitted-bank identities. Receipt hashes, frozen source hashes, the IV algebraic error decomposition, coverage indicators, and complete replication indices are checked before rendering.

Evidence identifiers

Base repository commit: b01dd19f22d44c3ca5dd6771772d1cb8f31fb755. Bridge identity: 9bb49b05fb87c356507ae5da0e0d8194fa29bb701f51c06f1e4cf23ea76359d7. Existing v6 identity: 8386d43620f5ad6b9df45cc88429245d708057e1a7d29ca90cce4d0bba8a8a5a. Detailed physical hashes and machine-readable results are in verification.json, summary.json, and input_receipt_hashes.json.

Mac packaging metadata entries are hash-checked against the transmitted source archive and excluded from scientific-source counts. They are not executable model modules. Original experiment chronology, archived source, raw records, and the original group PDF are preserved; no experiment is backdated.

Completion checklist

- Restored Tables 1-6 and Figures 1-6 in the original report's visual style.

- Restored learner-choice comparison with correctly labelled MC sampling spread.

- Corrected structural g versus outcome ell, IV estimator, SE, and histogram definitions.

- Separated the two baseline DGPs according to archived code; disclosed controlled learner priors.

- Completed the n=500/R=300 server-only bridge; reused n=300/R=200 raw evidence.

- Retained shared-bank controls, Series OLS / tree mechanisms, sensitivity extensions, and negative findings.